

LOW STYRENE SBR: PATENT LANDSCAPE FOR CONTROLLED STYRENE INCORPORATION, HYDROGENATION, AND TIRE-COMPOUND PERFORMANCE

1. ABSTRACT

This report surveys patents relevant to low styrene styrene-butadiene rubber (SBR) and closely related styrene-containing diene elastomers used in tire compounds. The target is a styrene-butadiene copolymer with reduced styrene incorporation relative to conventional SBR grades, typically achieved through controlled monomer feed, random copolymerization, solution polymerization, hydrogenation, or functionalized anionic initiation. Across the selected patents, the dominant technical themes are not only lowering or tuning styrene content, but also preserving wet grip, abrasion resistance, rolling resistance, processability, and filler compatibility in silica- and carbon-black-filled formulations.

The landscape includes patents that directly synthesize SBR with explicit styrene/butadiene feed ratios and hydrogenation levels, as well as patents that use low molecular weight styrene-butadiene copolymers, hydrogenated SBR, or styrene-containing multicomponent copolymers as performance modifiers. Several disclosures emphasize solution polymerization in hydrocarbon solvent with n-butyllithium and polar modifiers such as THF or TMEDA, followed by termination, hydrogenation, or functional end-group introduction. Other patents focus on post-polymerization compounding strategies: silica, silane coupling agents, liquid resins, and sulfur are used to balance traction, abrasion, and processability. Functional initiator patents show how sulfur-containing lithio initiators can generate SBR with tailored headgroups that improve filler interaction and reduce hysteresis.

Overall, the evidence indicates that low styrene SBR performance is best achieved by combining moderate-to-low styrene content with controlled molecular weight, narrow-to-moderate dispersity, and either hydrogenation or functionalization to recover reinforcement lost by reducing styrene. In tire compounds, the most consistent formulation trends are silica/carbon black hybrid reinforcement, silane coupling, and careful control of oil/resin loading and curing package. The patents collectively support a design space centered on solution-polymerized SBR with styrene content around the lower conventional range, hydrogenation where needed for durability, and filler-compatible end-group chemistry for improved compound performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US10392497B2
- Patent Title: Rubber compositions including siliceous fillers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is primarily directed to rubber compositions and filler coupling rather than SBR synthesis. It discloses blending an ethylenically unsaturated polymer bearing terminal or pendent aryl hydroxyl functionality with silica-containing filler particles or with a coupling compound that can bond the polymer to silica.

Relevant Experimental Evidence - No numbered synthesis examples were detected by the parser; the disclosure is formulation- and mechanism-focused. - The patent emphasizes covalent bonding between hydroxyaryl-functional polymers and silica surfaces to improve filler dispersion. - Functional initiators and terminal modification concepts are discussed as routes to prepare the polymer architecture.

Technical Relevance Relevant as a downstream compounding patent for low styrene SBR, because reduced-styrene polymers often need stronger filler coupling to recover reinforcement and hysteresis performance. It provides a silica-interaction strategy that can complement low styrene content in tire compounds.

Patent 2

- Patent Number: US11021595B2
- Patent Title: Multicomponent copolymer, rubber composition, cross-linked rubber composition, ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses multicomponent copolymers containing conjugated diene, non-conjugated olefin, and aromatic vinyl units, including styrene-containing tercopolymers. The polymers are used in rubber compositions and tires to balance high-temperature fracture performance with weather resistance. - Aromatic vinyl unit content: 3 mol % or more — Claim 1 - Non-conjugated olefin unit content: more than 44 mol % and less than 97 mol % — Claim 1 - Polystyrene-equivalent weight-average molecular weight: 100,000 to 9,000,000 — Claim 1 - Aromatic vinyl unit content: 30 mol % or less — Claim 2 - Conjugated diene unit content: 1 mol % or more — Claim 3 - Conjugated diene unit content: 40 mol % or less — Claim 4 - Ethylene unit alone as non-conjugated olefin unit — Claim 5 - Styrene unit contained in aromatic vinyl unit — Claim 6 - 1,3-butadiene unit and/or isoprene unit in conjugated diene unit — Claim 7 - Ter copolymer composed of 1,3-butadiene unit, ethylene unit and styrene unit — Claim 8 - Cis-1,4-bond content in conjugated diene unit: 50% or more — Claim 9 - Vulcanization temperature: 160 degrees C — Examples 1 to 9 - Vulcanization time: 30 minutes — Examples 1 to 9 - Silica: 50 parts by mass — Table 2 - Silane coupling agent: 5 parts by mass — Table 2 - Aromatic oil: 10 parts by mass — Table 2 - Stearic acid: 2 parts by mass — Table 2 - Antiaging agent: 0.5 parts by mass — Table 2 - Zinc oxide: 3 parts by mass — Table 2 - DPG accelerator: 1 part by mass — Table 2 - CZ accelerator: 1 part by mass — Table 2 - DM accelerator: 1 part by mass — Table 2 - Sulfur: 1.4 parts by mass — Table 2 - Ozone test temperature: 40 degrees C — Ozone resistance test - Ozone concentration: 50 pphm — Ozone resistance test - Static elongation: 30% — Ozone resistance test - Ozone exposure time: 48 hours — Ozone resistance test - High-temperature tensile test temperature: 100 degrees C — High-temperature fracture performance

Relevant Experimental Evidence - Example 1: multicomponent copolymers gave higher high-temperature fracture performance than the SBR control while maintaining no ozone cracking. - Examples 2-9: varying copolymer composition and blend ratio changed fracture indices while preserving ozone resistance. - Comparative Example 1: conventional SBR showed cracking under ozone exposure, highlighting the benefit of the multicomponent copolymer approach.

Technical Relevance This patent is relevant because it explicitly includes styrene-containing copolymers with reduced or controlled aromatic vinyl content, which overlaps the low styrene design space. It also shows how composition tuning can preserve tire durability when styrene content is constrained.

Patent 3

- Patent Number: US7462677B2
- Patent Title: Sulfur containing initiators for anionic polymerization of monomers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2008
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses sulfur-containing anionic initiators, especially lithio dithianes and oxathianes, for anionic polymerization of butadiene and butadiene/styrene. The initiators provide functional headgroups on SBR and related elastomers, improving filler interaction and lowering hysteresis in tire compounds. -

2-phenyl-1,3-dithiane: 2.1 g — Example 13 - 2-phenyl-1,3-dithiane: 10.69 mmol — Example 13 - THF: 5 mL — Example 13 - Cyclohexane: 10 mL — Example 13 - n-BuLi: 6.37 mL of 1.68 M in hexane — Example 13 - Initiator preparation temperature: -78 degrees C — Example 13 - Initiator stirring time: 3 hours — Example 13 - Initiator standing temperature: 0 degrees C — Example 13 - POT-Li concentration: 0.39 M — Example 15 - 2-phenyl-1,3-oxathiane: 1.0 g — Example 15 - 2-phenyl-1,3-oxathiane: 5.5 mmol — Example 15 - THF: 5.8 mL — Example 15 - Hexane: 5 mL — Example 15 - n-BuLi: 3.3 mL of 1.68 M in hexane — Example 15 - Initiator preparation temperature: -78 degrees C — Example 15 - Initiator stirring time: 3 hours — Example 15 - Initiator standing temperature: -5 degrees C — Example 15 - PDT-Li concentration: 0.5 M — Example 13 - Polymerization temperature: 24 degrees C — Example 35 - Polymerization temperature: 50 degrees C — Example 35 - Polymerization time: 16 minutes — Example 35 - Termination time: 30 minutes — Example 35 - Polymerization temperature: 48.9 degrees C — Example 45 - Polymerization time: 22 minutes — Example 45 - Additional reaction time: 41 minutes — Example 45 - Styrene content: 20.2% — Example 45 - Block styrene: 2.2% — Example 45 - 1,2-butadiene: 44.9% — Example 45 - Styrene content: 20.6% — Example 46 - Block styrene: 2.0% — Example 46 - 1,2-butadiene: 45.6% — Example 46 - Styrene content: 20.7% — Example 47 - Block styrene: 1.9% — Example 47 - 1,2-butadiene: 45.5% — Example 47 - Styrene content: 21.0% — Example 48 - Block styrene: 2.0% — Example 48 - 1,2-butadiene: 45.6% — Example 48 - Carbon black: 50 phr — Example 12 - Silica: 75 phr — Example 12 - Silane coupling agent A (Si69): 6 phr — Example 12 - Sulfur: 1.3 phr — Example 12 - Accelerators: 1.2 phr — Example 12 - Zinc oxide: 2 phr — Example 12 - Masterbatch temperature: 133 degrees C — Example 12 - Final mixer temperature: 65 degrees C — Example 12 - Final removal temperature: 100 to 105 degrees C — Example 12 - Cure temperature: 171 degrees C — Example 12

Relevant Experimental Evidence - Example 10: functionalized polymers reduced tan delta and delta G' versus control, indicating lower hysteresis. - Example 13: 2-lithio-2-phenyl-1,3-dithiane was synthesized and used as an anionic initiator. - Example 15: 2-lithio-2-phenyl-1,3-oxathiane was prepared and used to initiate butadiene/styrene polymerization. - Example 35: in-situ dithiane initiation plus DMI termination produced SBR with lower Mooney viscosity than n-BuLi control. - Example 45: low-styrene SBR around 20-21 wt% styrene was prepared with functional initiator chemistry and showed useful compound performance.

Technical Relevance This patent is highly relevant because it directly addresses anionic polymerization of butadiene/styrene with functional initiators, a key route for low styrene SBR. The disclosed styrene contents around 20 wt% and the performance gains in filled compounds make it especially pertinent to low-styrene tire-grade SBR.

Patent 4

- Patent Number: US10907034B2
- Patent Title: Pneumatic tire
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution polymerization of styrene and butadiene in n-hexane using n-butyllithium and THF, followed by termination and optional hydrogenation to produce SBR or hydrogenated SBR. The hydrogenated copolymer is then compounded into tire formulations with silica, carbon black, and silane coupling agents. - n-Hexane: 2,000 mL — Synthesis Example 1 - Styrene: 60 g — Synthesis Example 1 - Butadiene: 140 g — Synthesis Example 1 - THF: 2.5 g — Synthesis Example 1 - n-Butyllithium: 0.45 mmol — Synthesis Example 1 - Polymerization temperature: 50 degrees C — Synthesis Example 1 - Polymerization time: 5 hours — Synthesis Example 1 - Hydrogenation pressure: 0.4 MPa gauge — Synthesis Example 2 - Hydrogenation catalyst: titanocene dichloride-based catalyst — Synthesis Example 2 - Hydrogenation pressure: 0.7 MPa gauge — Synthesis Example 2 - Hydrogenation temperature: 90 degrees C — Synthesis Example 2 - Degree of hydrogenation: 96 mol % — Table 1 - Degree of hydrogenation: 77 mol % — Table 1 - Degree of hydrogenation: 99 mol % — Table 1 - Degree of hydrogenation: 80 mol % — Table 1 - Styrene content: 31% by mass — Table 1 - Styrene content: 30% by mass — Table 1 - Styrene content: 16% by mass — Table 1 - Styrene content: 20% by mass — Table 1 - Weight average molecular weight: 470,000 — Table 1 - Weight average molecular weight: 510,000 — Table 1 - Weight average molecular weight: 440,000 — Table 1 - Weight average molecular weight: 150,000 — Table 1 - Weight average molecular weight: 90,000 — Table 1 - Ethylene content: 67% by mass — Table 1 - Ethylene content: 48% by mass — Table 1 - Ethylene content: 19% by mass — Table 1 - Ethylene content: 64% by mass — Table 1 - Banbury mixing temperature: 150 degrees C — Examples - Banbury mixing time: 5 minutes — Examples - Open roll mixing temperature: 80 degrees C — Examples - Open roll mixing time: 5 minutes — Examples - Press vulcanization: 170 degrees C for 20 minutes — Examples - Die thickness: 0.5 mm — Examples

Relevant Experimental Evidence - Synthesis Example 1: conventional SBR was prepared by solution polymerization of styrene and butadiene. - Synthesis Example 2: hydrogenated SBR was produced using a titanocene dichloride catalyst under hydrogen pressure. - Synthesis Example 3: a higher hydrogenation target was achieved while maintaining similar molecular weight. - Synthesis Example 4: very high hydrogenation (95 mol %) was demonstrated. - Synthesis Example 5: amine-modified hydrogenated SBR was prepared, showing the role of end-group modification.

Technical Relevance This patent is directly relevant because it provides a clear solution-polymerized SBR synthesis route and shows how hydrogenation can tune ethylene

content and performance. Although the example SBR is not especially low in styrene, the disclosed process and composition ranges are useful for designing low styrene hydrogenated SBR grades.

Patent 5

- Patent Number: US7253235B2
- Patent Title: Rubber composition for a tire and tire using the same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2007
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low molecular weight styrene-butadiene copolymers prepared by solution polymerization in cyclohexane or hexane using n-butyllithium, with THF as a polar modifier in some examples. It also discloses olefin-conjugated diene-aromatic vinyl copolymers made by ethylene-containing polymerization and their use in tire compounds. - Cyclohexane: 1200 g — Preparation Example 1 - THF: 5 g — Preparation Example 1 - 1,3-Butadiene: 80 g — Preparation Example 1 - Styrene: 25 g — Preparation Example 1 - Temperature: 25 degrees C — Preparation Example 1 - n-Butyl lithium: 0.05 g — Preparation Example 1 - Polymerization time: 10 minutes — Preparation Example 1 - Styrene content: 25% by weight — Preparation Example 1 - Styrene content: 40% by weight — Preparation Example 2 - Styrene content: 5% by weight — Preparation Example 3 - Styrene content: 12% by weight — Preparation Example 4 - Weight average molecular weight: 10000 — Preparation Example 1 - Weight average molecular weight: 8000 — Preparation Example 2 - Weight average molecular weight: 14000 — Preparation Example 3 - Weight average molecular weight: 83000 — Preparation Example 4 - Toluene: 15 mL — Preparation Example 5 - Cyclopentadienyl triphenoxy titanium and MAO solution: 4 mL — Preparation Example 5 - [Al]/[Ti]: 500 — Preparation Example 5 - [Al]: 3.4 mol/L — Preparation Example 5 - Styrene: 4 mL — Preparation Example 5 - Butadiene: 4 mL — Preparation Example 5 - Ethylene partial pressure: 1.2 kg/cm² — Preparation Example 5 - Polymerization time: 1 hour — Preparation Example 5 - Cyclohexane: 15 mL — Preparation Example 8 - Butadiene: 2 mL — Preparation Example 8 - Styrene: 1 mL — Preparation Example 8 - THF: 0.5 mL — Preparation Example 8 - n-Butyl lithium: 0.3 mmol — Preparation Example 8 - Polymerization time: 5 minutes — Preparation Example 8 - Polymerization temperature: 40 degrees C — Preparation Example 8 - Mw: 2500 — Preparation Example 8 - Mw/Mn: 1.34 — Preparation Example 8 - Ethylene:butadiene:styrene ratio: 21:65:14 — Preparation Example 8 - Mw: 5000 — Preparation Example 6 - Mw/Mn: 1.42 — Preparation Example 6 - Ethylene:butadiene:styrene ratio: 20:67:13 — Preparation Example 6 - Mw: 500 — Preparation Example 7 - Mw/Mn: 1.25 — Preparation Example 7 - Ethylene:butadiene:styrene ratio: 18:68:14 — Preparation Example 7 - Butadiene:styrene ratio: 85:15 — Preparation Example 8 - Press vulcanization temperature: 170 degrees C — Examples - Press vulcanization time: 20

Relevant Experimental Evidence - Preparation Example 1: low molecular weight styrene-butadiene copolymers with styrene contents from 5 to 40 wt% were synthesized. - Preparation Example 5: ethylene-containing olefin-conjugated diene-aromatic vinyl copolymers were prepared under ethylene pressure. - Preparation Example 8: a low molecular weight styrene-butadiene copolymer with 15% styrene was used as a comparative material. - Examples 15-17: tire compounds using the olefin-conjugated diene-aromatic vinyl copolymers were evaluated for grip and abrasion. - The patent shows that low molecular weight styrene-butadiene copolymers can be used as tire additives to tune performance.

Technical Relevance This patent is relevant because it explicitly discloses low styrene styrene-butadiene copolymers, including a 5 wt% styrene example. It also provides a route for using such low molecular weight copolymers in tire compounds to balance grip and abrasion resistance.

Patent 6

- Patent Number: US10759885B2
- Patent Title: Hydrogenated branched conjugated diene copolymer, rubber composition and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenated branched conjugated diene copolymers made by copolymerizing branched dienes such as myrcene or farnesene with styrene and optionally butadiene/isoprene, then hydrogenating the resulting copolymer. The polymers are compounded with silica, carbon black, and silane coupling agents to improve tire performance and processability. - Hexane: 2000 mL — Preparation Example 1 - Butadiene: 110 g — Preparation Example 1 - Styrene: 90 g — Preparation Example 1 - TMEDA: 0.22 mmol — Preparation Example 1 - n-Butyllithium: 35 mmol — Preparation Example 1 - Polymerization temperature: 50 degrees C — Preparation Example 1 - Polymerization time: 5 hours — Preparation Example 1 - Hydrogenation pressure: 5.0 kgf/cm² — Preparation Example 2 - Hydrogenation temperature: 80 degrees C — Preparation Example 2 - Degree of hydrogenation: 50% — Preparation Example 2 - Butadiene: 150 g — Preparation Example 3 - Styrene: 125 g — Preparation Example 3 - Isoprene: 95 g — Preparation Example 9 - Myrcene: 150 g — Preparation Example 5 - Farnesene: 150 g — Preparation Example 15 - Farnesene: 55 g — Preparation Example 17 - Butadiene: 95 g — Preparation Example 17 - Styrene: 125 g — Preparation Example 17 - Styrene: 125 g — Preparation Example 7 - Styrene: 125 g — Preparation Example 13 - Styrene: 125 g — Preparation Example 19 - Carbon black: 5 phr — Tables 2 to 4 - Silica: 75 phr — Tables 2 to 4 - Silane coupling agent A (Si69): 6.0 phr — Tables 2

to 4 - Silane coupling agent B (NXT): 0.8 to 7.5 phr — Tables 2 to 4 - Silane coupling agent C (NXT-Z45): 6.0 phr — Tables 2 to 4 - Antioxidant: 1.5 phr — Tables 2 to 4 - Stearic acid: 2 phr — Tables 2 to 4 - Zinc oxide: 2.5 phr — Tables 2 to 4 - Wax: 1 phr — Tables 2 to 4 - Sulfur: 2 phr — Tables 2 to 4 - Accelerator (1): 1.8 phr — Tables 2 to 4 - Accelerator (2): 1.2 phr — Tables 2 to 4

Relevant Experimental Evidence - Preparation Example 1: a styrene-butadiene copolymer was synthesized by solution polymerization with TMEDA and n-BuLi. - Preparation Example 2: the copolymer was hydrogenated at 5.0 kgf/cm² and 80 degrees C to 50% hydrogenation. - Preparation Example 5: a styrene-butadiene copolymer containing an amine modifier was prepared and hydrogenated. - Preparation Example 15: a farnesene-styrene copolymer was synthesized, showing branched diene incorporation. - Examples 1-16: silica-filled compounds with hydrogenated copolymers and silane coupling agents showed improved fuel economy, tensile strength, and abrasion resistance.

Technical Relevance This patent is relevant because it uses styrene-butadiene copolymers and hydrogenation to tune tire performance, while also showing how branched diene incorporation can modify processability. It supports the broader low styrene SBR design space by demonstrating controlled styrene-containing copolymer synthesis and compounding.

Patent 7

- Patent Number: US7829624B2
- Patent Title: One-pot synthesis of nanoparticles and liquid polymer for rubber applications
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-batch synthesis of liquid polymer and in situ-formed nanoparticles in hydrocarbon solvent. The liquid polymer can be styrene-butadiene copolymer, and the process uses partial termination followed by addition of mono-vinyl aromatic and multiple-vinyl aromatic monomers to form core-shell nanoparticles in the same reactor. - Partial termination: 20 to 95% of polymers terminated — Claim 2 - Reaction temperature: 20 to 140 degrees C — Process description - Crosslinking density: 0.01 to 1 — Process description - Crosslinking density: 0.1 to 0.8 — Process description - Crosslinking density: 0.2 to 0.4 — Process description - Crosslinking density: 0.3 — Process description - Crosslinking density: 0.8 to 1.0 — Process description - Crosslinking density: 0.9 — Process description - Liquid polymerization termination time: about 15 minutes — Process description - Liquid polymerization termination range: 1 to 99 wt% terminated and 99 to 1 wt% unteminated — Process description - Preferred terminated fraction: 25 to 75 wt% — Process description

Relevant Experimental Evidence - The patent explains that the liquid polymer can be a styrene-butadiene copolymer formed in hydrocarbon solvent. - Partial termination is used to leave living chains for nanoparticle shell formation. - The in situ nanoparticle step crosslinks micelles using multiple-vinyl aromatic monomers. - The blend is intended to improve dispersion and processing versus separately made nanoparticles and liquid polymer.

Technical Relevance This patent is relevant because it provides a route to styrene-butadiene liquid polymers and in situ nanoparticle blends that can be used in tire compounds. While not a low styrene SBR synthesis patent per se, it is useful for compounding and processability strategies around SBR-based systems.

Patent 8

- Patent Number: US10428203B2
- Patent Title: Pneumatic tire
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution polymerization of styrene and butadiene in n-hexane using n-butyllithium and TMEDA, followed by optional hydrogenation to produce hydrogenated SBR. The hydrogenated copolymer is then compounded with silica, carbon black, and silane coupling agents for tire applications. - n-Hexane: 2,000 mL — Synthesis Example 1 - Styrene: 60 g — Synthesis Example 1 - Butadiene: 140 g — Synthesis Example 1 - TMEDA: 0.93 g — Synthesis Example 1 - n-Butyllithium: 0.45 mmol — Synthesis Example 1 - Polymerization temperature: 50 degrees C — Synthesis Example 1 - Polymerization time: 5 hours — Synthesis Example 1 - Hydrogenation pressure: 0.4 MPa gauge — Synthesis Example 2 - Hydrogenation catalyst: titanocene dichloride-based catalyst — Synthesis Example 2 - Hydrogenation pressure: 0.7 MPa gauge — Synthesis Example 2 - Hydrogenation temperature: 90 degrees C — Synthesis Example 2 - Degree of hydrogenation: 60 mol % — Synthesis Example 2 - Degree of hydrogenation: 80 mol % — Synthesis Example 3 - Degree of hydrogenation: 95 mol % — Synthesis Example 4 - Degree of hydrogenation: 95 mol % — Synthesis Example 5 - Styrene content: 30% by mass — Table 1 - Butadiene content: 70% by mass — Table 1 - Weight average molecular weight: 490,000 — Table 1 - Weight average molecular weight: 450,000 — Table 1 - Weight average molecular weight: 480,000 — Table 1 - Weight average molecular weight: 460,000 — Table 1 - Glass transition temperature: -30 degrees C — Table 1 - Glass transition temperature: -31 degrees C — Table 1 - Silica: 75 phr — Tables 2 to 4 - Carbon black: 5 phr — Tables 2 to 4 - Oil: 25 phr — Tables 2 to 4 - Silane coupling agent A (Si69): 6.0 phr — Tables 2 to 4 - Silane coupling agent B (NXT): 0.8 to 7.5 phr — Tables 2 to 4 - Silane coupling agent C (NXT-Z45): 6.0 phr — Tables 2 to 4 - Sulfur: 2 phr — Tables 2 to 4 - Accelerator (1): 1.8 phr — Tables 2 to 4 - Accelerator (2): 1.2 phr — Tables 2 to 4

Relevant Experimental Evidence - Synthesis Example 1: a 30 wt% styrene SBR was prepared by solution polymerization. - Synthesis Example 2: the SBR was hydrogenated to 60 mol % using a titanocene catalyst. - Synthesis Example 3: hydrogenation was increased to 80 mol %. - Synthesis Example 4: hydrogenation was increased to 95 mol %. - Examples 5 and the compound tables show that hydrogenated modified SBR can be used in silica-filled tire compounds.

Technical Relevance This patent is relevant because it provides a direct solution-polymerized SBR synthesis route with explicit styrene content and hydrogenation control. It is useful for low styrene SBR development because it demonstrates how styrene level, hydrogenation, and filler package interact in tire compounds.

Patent 9

- Patent Number: US8993664B2
- Patent Title: Rubber composition for tires and pneumatic tire
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is a tire-compound formulation disclosure rather than an SBR synthesis patent. It combines diene rubber with fine silica, sulfur, and a liquid resin, and optionally aromatic vinyl polymer resin, to improve handling stability, fuel economy, wet grip, elongation at break, and abrasion resistance. - Silica BET specific surface area: 170 to 270 m²/g — Claim 1 - Liquid resin softening point: -20 to 45 degrees C — Claim 1 - Liquid resin amount: 0.5 to 20 parts by mass — Claim 1 - Silica amount: 40 to 120 parts by mass — Claim 1 - Aromatic vinyl polymer softening point: 46 to 140 degrees C — Claim 1 - Combined amount of oil, liquid resin, and aromatic vinyl polymer: 6 to 30 parts by mass — Claim 1 - Silica BET specific surface area: 190 to 250 m²/g — Claim 2 - Zinc oxide: 0.5 to 2.9 parts by mass — Claim 3 - Silica BET specific surface area: 210 to 250 m²/g — Claim 4 - Carbon black nitrogen adsorption specific surface area: 90 to 130 m²/g — Claim 5

Relevant Experimental Evidence - No numbered synthesis examples were detected by the parser; the disclosure is formulation-focused. - The patent emphasizes liquid coumarone-indene resin and aromatic vinyl polymer resin as dispersion aids. - It identifies fine silica and sulfur dispersion as key issues in diene-rubber tire compounds. - The patent is aimed at balancing handling stability, fuel economy, wet grip, elongation, and abrasion resistance.

Technical Relevance This patent is relevant as a compounding strategy for low styrene SBR, especially where reduced styrene content may require stronger filler and resin-based reinforcement. It does not synthesize SBR, but it provides a formulation framework that can help low styrene SBR compounds maintain tire performance.

Patent 10

- Patent Number: US6057397A
- Patent Title: Rubber composition and process for preparing the same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2000
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses silica-filled diene rubber compositions using hydroxyl group-containing diene rubber prepared by copolymerization or by post-polymerization modification of a diene polymer with bound active metal. The process emphasizes staged silica addition to improve dispersion and processability. - Weight-average molecular weight: 50,000 or more — Claim 1 - Conjugated diene monomer content: 40 to 100% by weight — Claim 3 - Copolymerizable monomer content: 0 to 60% by weight — Claim 3 - Hydroxyl group-containing diene rubber (A) introduced by modifier reaction — Claim 1 - Hydroxyl group-containing diene rubber (A) introduced by ketone, ester, aldehyde, or epoxy — Claim 2

Relevant Experimental Evidence - No numbered synthesis examples were detected by the parser; the disclosure is process- and composition-focused. - The patent teaches hydroxyl-functional diene rubber for better silica reinforcement. - It also teaches staged mixing: first combine diene rubber with part of the silica, then add the remainder and other compounding agents. - The disclosure is aimed at improving resilience, tensile strength, abrasion resistance, hardness, and processability.

Technical Relevance This patent is relevant because hydroxyl-functional diene rubber is a classic route to improve silica compatibility in SBR-based tire compounds. For low styrene SBR, such functionalization can compensate for reduced aromatic content and preserve reinforcement and processability.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US10907034B2 and US10428203B2 both disclose solution-polymerized SBR using n-hexane, n-butyllithium, and THF/TMEDA, with styrene at 60 g and butadiene at 140 g in the main synthesis examples; both then use hydrogenation to tune performance.
- US10907034B2 reports a 30 wt% styrene SBR and hydrogenated variants with 16-30 wt% styrene-equivalent composition, while US10428203B2 reports a 30 wt% styrene SBR and hydrogenation levels of 60-95 mol %; both sit at the upper end of conventional SBR styrene content rather than the low-styrene target.

- US7253235B2 is the clearest low-styrene disclosure, with preparation examples at 5 wt%, 12 wt%, 25 wt%, and 40 wt% styrene and low molecular weights from 8,000 to 83,000, making it more directly aligned with low-styrene SBR than the hydrogenated tire patents.
- US7462677B2 and US6057397A focus on functionality and silica compatibility rather than styrene reduction itself; both are relevant because low styrene SBR often needs end-group or hydroxyl functionality to recover filler reinforcement.
- US10759885B2 and US11021595B2 broaden the styrene-containing elastomer space by combining styrene with branched dienes or ethylene/non-conjugated olefin units; these are more about performance balancing than strict low-styrene SBR, but they show how styrene can be diluted or structurally offset.
- US8993664B2 and US10392497B2 are compounding patents: the former uses liquid resin plus fine silica and sulfur, while the latter uses hydroxyaryl-functional polymer/silica coupling. Both are important downstream tools for low styrene SBR formulations.
- US7829624B2 adds a process route for one-pot liquid polymer/nanoparticle blends, useful for tire compounds containing SBR-based liquid polymers, but it does not directly optimize styrene content.
- Across the set, the most common tire-compound reinforcement package is silica plus silane coupling agent, often with carbon black and oil/resin; hydrogenation and functionalization are the main polymer-level strategies used to offset performance losses from lower styrene content.

4. CONCLUSION

The patent evidence supports low styrene SBR development through three complementary strategies: direct solution polymerization with controlled styrene feed, hydrogenation or structural modification to recover durability, and filler-compatible functionalization to preserve reinforcement. The most directly relevant low-styrene disclosure is US7253235B2, which includes styrene contents as low as 5 wt% in low molecular weight styrene-butadiene copolymers. For tire-grade systems, the strongest supporting process conditions are solution polymerization in hydrocarbon solvent with n-butyllithium and a polar modifier such as THF or TMEDA, followed by careful termination and, where needed, hydrogenation or end-group modification. On the formulation side, silica/silane coupling, liquid resin assistance, and hydroxyl- or sulfur-functional polymer architectures are repeatedly shown to improve wet grip, abrasion resistance, and processability. A practical evidence-based design window for low styrene SBR is therefore a solution-polymerized styrene-butadiene copolymer in the lower conventional styrene range, combined with silica/silane reinforcement and either hydrogenation or functional end-group chemistry to maintain tire performance.

5. REFERENCES

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- US10759885B2 | Hydrogenated branched conjugated diene copolymer, rubber composition and ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10759885B2/en>
- US7829624B2 | One-pot synthesis of nanoparticles and liquid polymer for rubber applications | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7829624B2/en>
- US10428203B2 | Pneumatic tire | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10428203B2/en>
- US8993664B2 | Rubber composition for tires and pneumatic tire | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8993664B2/en>
- US6057397A | Rubber composition and process for preparing the same | Not disclosed | US | 2000 | <https://patents.google.com/patent/US6057397A/en>